1 Auswertung

Die mittlere Temperatur ist $\bar{T}$ (als inline-Formula). Das ist auch eine in den Text eingebetete Formula: $f(x) = 2.0 \times x^2 + 45 \times x$. Hier is auch ein gutes Beisoiel für eingebetete Formel:$\sqrt{x^2+1}$ ....

$$\bar{T} = \frac{1}{n} \sum_{i=1}^{n} T_i \tag{1}$$

Die Berechnung erfolgt mit Gleichung (1).

Mehrere Umformungen werden ausgerichtet:

$$\begin{aligned} \Delta T &= T_{\text{Start}} - T_{\text{Ende}} \\ &= 65{,}0 - 44{,}1 = 20{,}9. \end{aligned}$$

Beispiel: Generalisierte gyroskopische Kräfte:

$$\begin{aligned} K_0 = & \\ & - m_1 l_1 L_1 \dot{\theta_1}^2 \cos(\theta_0 - \theta_1) \\ & + m_2 l_2 L_2 \dot{\theta_2}^2 \cos(\theta_0 - \theta_2) \end{aligned} \tag{2}$$

$$\begin{aligned} K_1 = & \\ & m_1 L_1 l_1 \dot{\theta_0}^2 \cos(\theta_0 - \theta_1) \end{aligned} \tag{3}$$

$$\begin{aligned} K_2 = & \\ & - m_2 L_2 l_2 \dot{\theta_0}^2 \cos(\theta_0 - \theta_2) \end{aligned} \tag{4}$$

Lagrangian:

$$L = \sum_{i=1}^{n} T - \sum_{i=1}^{n} U$$

Matrix

$$\begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \tag{5}$$

Quadratic Equation #1

$$\begin{cases} y = a + bx \\ R^2 = (x - x_0)^2 + (y - y_0)^2 \\ P^2 = x_0^2 + y_0^2 \end{cases}$$

$$R^2 = x_0^2 - 2xx_0 + x^2 + y_0^2 - 2yy_0 + y^2$$

$$x^2(1 + b^2) - x2(x_0 + b\sqrt{P^2 - x_0^2} - ab) + (P^2 + a^2 - R^2 - 2a\sqrt{P^2 - x_0^2}) = 0$$

$$D = (2(x_0 - b\sqrt{P^2 - x_0^2} - ab))^2 - 4(1 + b^2)(P^2 + a^2 - R^2 - 2a\sqrt{P^2 - x_0^2})$$

Another way, which is considered to be much better:

$$\begin{cases} S^2 = (x - X_S)^2 + (y - X_S)^2 \\ R^2 = (x - x_0)^2 + (y - y_0)^2 \\ P^2 = (x_0 - 0)^2 + (y_0 - 0)^2 \end{cases}$$

$$\begin{cases} S^2 = (x - x_S)^2 + (y - x_S)^2 \\ R^2 = (x - x_R)^2 + (y - y_R)^2 \\ P^2 = (x_R - x_P)^2 + (y_R - y_P)^2 \end{cases}$$

Quadratic Equation #2

$$\begin{cases} y = a + bx \\ R^2 = (x - x_0)^2 + (y - y_0)^2 \end{cases}$$

$$R^2 = (x - x_0)^2 + ((a + bx) - y_0)^2$$

$$R^2 = (x - x_0)^2 + (bx + (a - y_0))^2$$

$$x^2 - 2x_0x + x_0^2 + b^2x^2 + 2b(a - y_0)x + (a - y_0)^2 - R^2 = 0$$

$$x^2(1 + b^2) + x(2b(a - y_0) - 2x_0) + (x_0^2 + (a - y_0)^2 - R^2) = 0$$

$$D = (2(ab - by_0 - x_0))^2 - 4(1 + b^2)(x_0^2 + (y_0 - a)^2 - R^2)$$

$$x_1 = \frac{-2(ab-by_0-x_0)+\sqrt{D}}{2(1+b^2)}$$
$$x_2 = \frac{-2(ab-by_0-x_0)-\sqrt{D}}{2(1+b^2)}$$

$$y = y = a + b(\max(x_1, x_2)) \implies y = y = a + bx_2$$

$$\begin{cases} x_1 = \frac{-2(ab-by_0-x_0)+\sqrt{(2(ab-by_0-x_0))^2-4(1+b^2)(x_0^2+(y_0-a)^2-R^2)}}{2(1+b^2)} \\ y = a + b(\frac{-2(ab-by_0-x_0)+\sqrt{(2(ab-by_0-x_0))^2-4(1+b^2)(x_0^2+(y_0-a)^2-R^2)}}{2(1+b^2)}) \end{cases}$$

$$x_0 = R\cos\alpha$$
$$y_0 = R\sin\alpha$$

$$\begin{cases} x_1 = \frac{-2(ab-b(R\sin\alpha)-R\cos\alpha)+\sqrt{(2(ab-b(R\sin\alpha)-R\cos\alpha))^2-4(1+b^2)((R\cos\alpha)^2+((R\sin\alpha)-a)^2-R^2)}}{2(1+b^2)} \\ y = a + b(\frac{-2(ab-b(R\sin\alpha)-(R\cos\alpha))+\sqrt{(2(ab-b(R\sin\alpha)-(R\cos\alpha)))^2-4(1+b^2)((R\cos\alpha)^2+((R\sin\alpha)-a)^2-R^2)}}{2(1+b^2)}) \end{cases}$$

$\frac{d}{dx}(\frac{1}{(1+b^2)}((R(bsin(x)+cos(x))(1-2a+2\sqrt{R^2(bsin(x)+cos(x))^2-2aR(bcos(x)+sin(x))+a^2})+$
$R^2(bsin(x)+cos(x))^2(2ab+1)-ab(ab+1)-2a^2))) =$

$= \frac{1}{(1+b^2)}$
$(R(bcos(x)-sin(x))(2\sqrt{a^2-2aR(bcos(x)+sin(x))+R^2(bsin(x)+cos(x))^2}-2a+1)+$
$\frac{R(bsin(x)+cos(x))(2R^2(bcos(x)-sin(x))(bsin(x)+cos(x))-2aR(cos(x)-bsin(x)))}{\sqrt{a^2-2aR(bcos(x)+sin(x))+R^2(bsin(x)+cos(x))^2}} +$
$2R^2(2ab+1)(bcos(x)-sin(x))(bsin(x)+cos(x)))$